

VILP -- SYSTEM ARCHITECTURE BLUEPRINT

Verified Internship Lifecycle Platform // Enterprise Technical Architecture Deck

- Core Mission: Institutional single-active mutex internship platform for university placement governance.
- 4-Way Stakeholder Triangulation: Students, Corporate Recruiters, Faculty Mentors, and T&P Head in one live loop.
- Tech Stack: React 18 SPA + Vite + Tailwind | Java 21 + Spring Boot 3.3.x | PostgreSQL 16 (PgBouncer) | Gemini 2.0 AI.
- Security Standard: Stateless JWT Token Rotation (Argon2id/BCrypt) + Deterministic RBAC Gate + XSS Request Sanitizer.
- Compliance: AICTE Section 7.2, NEP-2020 Accreditation, and 240-hour contact compliance monitoring.

1. HIGH-LEVEL 4-TIER ARCHITECTURE

Decoupled Presentation, Security Gateway, Domain Services, and Cloud Data Tier

- TIER 1: Presentation Layer (SPA) - React 18, Vite, Tailwind CSS, TanStack Query, Lucide React (Student, Recruiter, Mentor, T&P).
- TIER 2: Security & Gateway Layer - Sliding Window Rate Limiting, JwtAuthenticationFilter, InputSanitizationAdvice, Method RBAC.
- TIER 3: Backend Domain Core - Spring Boot 3.3 (AuthService, Rule Engine, NOC Workflow, Logbook Telemetry, QR Certificate Engine).
- TIER 4: Data & Cloud Layer - PostgreSQL 16 on Supabase with HikariCP (prepareThreshold=0), Firebase Auth, Cloud Storage.
- Inter-Service Decoupling: Fully stateless REST architecture with automated token rotation and sub-50ms query execution.

2. END-TO-END INTERNSHIP LIFECYCLE WORKFLOW

10-Stage Deterministic State Machine from Opportunity Posting to QR Verification

1. Opportunity Posting: Recruiter posts opening (DRAFT) -> T&P audits organization & publishes listing (PUBLISHED).
2. Deterministic Screening: Applicant CGPA, active backlogs, and department are checked instantly before application.
3. Application & Offer: 1-Click verified institutional apply -> Recruiter reviews Kanban pipeline & issues Digital Offer.
4. Automated Digital NOC: Student initiates request -> T&P multi-tier digital sign-off -> Assigns Faculty Mentor.
5. Weekly Telemetry & Completion: Student submits weekly work -> Mentor audits (1-5 Stars) -> Employer signs final review.
6. Cryptographic Certificate: T&P issues QR-code verifiable completion certificate with SHA-256 tamper-proof ledger hash.

3. DUAL-LAYER AUTHENTICATION & SECURITY FLOW

Firebase Google OAuth Identity Federation + Spring Security Stateless JWT

- Flow A (Google 1-Click): Client signs in via Firebase SDK -> Sends verified UID & ID Token to POST /api/auth/firebase-login.
- Auto-Provisioning Engine: Backend validates Firebase identity, assigns institutional role, and provisions verified profile.
- Flow B (Institutional Direct): Email/Password authentication verified against Argon2id / BCrypt hash with lock after 5 failures.
- Token Management: Issues short-lived Access Token (30 min) + secured Refresh Token (7 days) with automated silent rotation.
- RBAC Gatekeeper: @PreAuthorize enforces granular method security across 6 roles (STUDENT, COMPANY, MENTOR, TNP_OFFICER, TNP_HEAD, ADMIN).

4. DETERMINISTIC ACADEMIC RULE ENGINE

Algorithmic Pre-Screening Eliminating Manual Resume Overhead for Recruiters

- Rule 1 (CGPA Cutoff): Verified against institutional database (e.g. Student CGPA 8.42 $\geq$ Required 8.00).
- Rule 2 (Backlog Policy): Enforces strict maximum permissible active backlogs (e.g. Max Backlogs == 0).
- Rule 3 (Department Filter): Checks department ID code against target disciplines (CSE, IT, ECE, EE, ME, CE).
- Rule 4 (Graduation Cohort): Validates passing year batch eligibility (e.g. Batch == 2026).
- Boolean Conjunctive Evaluator: Evaluates R1 and R2 and R3 and R4 in $< 5\text{ms}$. Ineligible students receive AI skill gap advice.

5. MULTI-BRANCH ENGINEERING ECOSYSTEM

Full First-Class Support Across All 10+ University Engineering Disciplines

- Computer Science & IT (CSE/IT/MCA): GitHub commits, PR reviews, API endpoints, microservices, and system design docs.
- Mechanical & Automobile (ME/AE): SolidWorks/AutoCAD 3D CAD models, ANSYS FEA simulation reports, shop floor work orders.
- Civil Engineering (CE): Site survey photos, structural blueprints, concrete compressive strength test logs, AutoCAD drawings.
- Electrical & ECE (EE/ECE): Altium/Eagle PCB schematics, MATLAB/Simulink waveforms, embedded firmware C code, oscilloscope logs.
- Domain Mentor Matching: Mechanical students are mentored by Mechanical faculty; Civil students by Civil professors.

6. 5-LAYER ANTI-FRAUD & TELEMETRY ENGINE

Preventing Fake Certificates, Ghost Job Postings, and Fabricated Work Hours

- Layer 1 (Multi-Party Triangulation): Student + Company Manager + Faculty Mentor + T&P Cell must all co-sign.
- Layer 2 (Tangible Proof-of-Work): Logbooks mandate real artifacts (CAD renders, Git PRs, site logs, test reports).
- Layer 3 (Time-Gated Submission): Strict weekly calendar windows prevent end-of-semester bulk fake dumps.
- Layer 4 (Employer Performance Rating): Recruiter confirms physical attendance, stipend payout, and PPO offer.
- Layer 5 (Public QR Verification): Anyone can scan certificate QR code to inspect full verified audit trail on `/verify-certificate/{id}`.

7. ACTIVITY HEATMAP & TELEMETRY PIPELINE

112-Day Rolling GitHub/LeetCode Contribution Matrix for Continuous Student Accountability

- Rolling 16-Week Grid Matrix: 112 discrete daily cells (16 columns x 7 rows) synchronized with backend logbook reports.
- 5-Tier Intensity Color Coding: Level 0 (0h Rest), Level 1 (<4h), Level 2 (4-7h), Level 3 (7-9h Full Workday), Level 4 (>9h Sprint).
- Streak Engine: Calculates active daily streaks, longest historical streak, and total accredited hours against 240h target.
- Interactive Drilldown: Clicking any calendar day reveals verified tasks, deliverables, and mentor star grading (1-5 Stars).
- T&P Telemetry Dashboard: Placement officers inspect cohort heatmaps to spot inactive students before dropouts occur.